Static Visualization and Statistical Analysis

Group 10

Project - 1 Report

Course Code: IE6600
Course Name: Computation and Visualization for Analytics

Team Members:
Clyde Joseph Dsouza
Aditya Girish Anand
Sayee Ashish Aher

Introduction :

In the transition toward sustainable transportation, understanding which vehicle technologies deliver the best fuel economy and environmental performance is critical for consumers, fleet managers, and policymakers. This project analyzes the EPA SmartWay Vehicle List for Model Year 2025, containing 763 vehicle configurations across various fuel types, vehicle classes, and drivetrain options.

EPA's SmartWay program, launched in 2004, represents a voluntary public-private partnership designed to improve freight and passenger vehicle efficiency while reducing greenhouse gas emissions and air pollution. By certifying vehicles, carriers, and technologies that exceed standard performance thresholds, SmartWay provides market recognition for superior environmental outcomes. The MY 2025 dataset reflects cutting-edge automotive technologies including electric, hybrid, hydrogen, and advanced gasoline powertrains, making it an ideal source for evaluating next-generation vehicle efficiency patterns.

The primary objective is to identify how fuel type, vehicle class, drivetrain configuration, and SmartWay certification ratings relate to combined MPG and greenhouse gas performance. Using matplotlib for static visualizations and statistical methods including correlation analysis, ANOVA, t-tests, and linear regression, we examine patterns in fuel economy and emission scores to provide actionable insights for stakeholders evaluating vehicle efficiency.

This analysis addresses three key research questions: (1) Which fuel technologies deliver superior combined MPG and how do they compare statistically? (2) Do vehicle class and drivetrain choices significantly impact efficiency, and if so, how? (3) What is the quantitative relationship between fuel economy and greenhouse gas emissions in this high-performance vehicle subset? The structured approach data inspection, cleaning, exploratory visualization, statistical hypothesis testing, and regression modeling enables rigorous evaluation of these questions while demonstrating proficiency in static data visualization and probability/statistics fundamentals required by IE6600.

Tasks:

1. Data Selection and Confirmation:

For this project we used **SmartWay Vehicle List for MY 2025.xlsx** dataset from data.gov website. The "SmartWay Vehicle List for MY 2025" dataset was selected from the EPA's data catalog, providing comprehensive Model Year 2025 vehicle efficiency ratings across 763 configurations. This dataset contains critical variables including fuel type, vehicle class, drivetrain, MPG ratings, air pollution scores, and greenhouse gas scores, enabling analysis of next-generation sustainable transportation technologies.

Obtained approval from the teaching assistant to ensure our dataset is unique among the groups.

2. Data Acquisition and Inspection:

The SmartWay dataset contains 763 rows and 17 columns describing vehicle models certified under the EPA SmartWay program. Key variables include Model name, engine displacement (Displ), cylinders (Cyl), transmission (Trans), drivetrain (Drive), fuel type (Fuel), certification region (Cert Region), vehicle class (Veh Class), Air Pollution Score, City/Highway/Combined MPG, Greenhouse Gas Score, and SmartWay rating.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 763 entries, 0 to 762
Data columns (total 17 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Model                   763 non-null    object
 1   Displ                   102 non-null    float64
 2   Cyl                     102 non-null    float64
 3   Trans                   763 non-null    object
 4   Drive                   763 non-null    object
 5   Fuel                    763 non-null    object
 6   Cert Region             763 non-null    object
 7   Stnd                    763 non-null    object
 8   Stnd Description        763 non-null    object
 9   Underhood ID            763 non-null    object
 10  Veh Class               763 non-null    object
 11  Air Pollution Score     763 non-null    int64
 12  City MPG                763 non-null    object
 13  Hwy MPG                 763 non-null    object
 14  Cmb MPG                 763 non-null    object
 15  Greenhouse Gas Score    763 non-null    int64
 16  SmartWay                763 non-null    object
dtypes: float64(2), int64(2), object(13)
memory usage: 101.5+ KB
```

	0
Model	0
Displ	661
Cyl	661
Trans	0
Drive	0
Fuel	0
Cert Region	0
Stnd	0
Stnd Description	0
Underhood ID	0
Veh Class	0
Air Pollution Score	0
City MPG	0
Hwy MPG	0
Cmb MPG	0
Greenhouse Gas Score	0
SmartWay	0

dtype: int64

Fig.1: Dataset information summary

Initial inspection revealed City, Highway, and Combined MPG stored as object types and 62 rows with missing MPG values after numeric conversion. Fig.1, shows the cleaned dataset structure with 701 complete observations across 17 variables, providing sufficient sample size for robust statistical analysis.

3. Data Cleaning and Preparation:

Data cleaning transforms raw datasets into reliable analysis ready formats by addressing missing values, incorrect data types, and inconsistencies that could bias results. This preparation phase ensures subsequent statistical analysis and visualizations produce accurate, interpretable insights.

```python
# Step 3: Data Cleaning and Preparation

# 3.1 Handle MPG columns
# City MPG, Hwy MPG, and Cmb MPG are objects; convert them to numeric:
mpg_cols = ['City MPG', 'Hwy MPG', 'Cmb MPG']

for col in mpg_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')

df[mpg_cols].info()
df[mpg_cols].isna().sum()



# 1. Drop rows with missing MPG (for simplicity)
# Drop rows where any MPG value is missing
df = df.dropna(subset=mpg_cols)

# Quick check
df[mpg_cols].isna().sum()
len(df)
# You should see 0 missing in those three columns and around 701 rows remaining.
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 763 entries, 0 to 762
Data columns (total 3 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   City MPG  701 non-null    float64
 1   Hwy MPG   701 non-null    float64
 2   Cmb MPG   701 non-null    float64
dtypes: float64(3)
memory usage: 18.0 KB
701
```

Fig.2: Dataset Cleaning

Missing Data and Data Types:

In fig.2, City MPG, Hwy MPG, and Cmb MPG columns contained string values that were converted to numeric using pd.to_numeric(errors='coerce'). This conversion revealed 62 rows with missing MPG values, which were dropped to maintain complete fuel economy data for analysis. No duplicate rows were identified.

Categorical Variables:

Categorical variables (Fuel, Veh Class, Drive, SmartWay, Cert Region) were retained as object types for grouping operations in visualizations and ANOVA tests. Fuel types include Electricity, Gasoline, Hydrogen, and hybrids; vehicle classes span small cars to pickups; drivetrains include 2WD and 4WD configurations; SmartWay ratings include "Yes" and "Elite" designation.

4. Exploratory Data Analysis (EDA:

Exploratory Data Analysis (EDA) uses visualizations to reveal patterns, trends, and relationships before formal statistical testing, guiding hypothesis formation and identifying data characteristics that inform analysis strategy.

4.1. Distributions of Fuel Economy and Emission Scores:

Histograms summarize univariate distributions to reveal central tendency, spread, skewness, and outliers in key performance metrics.

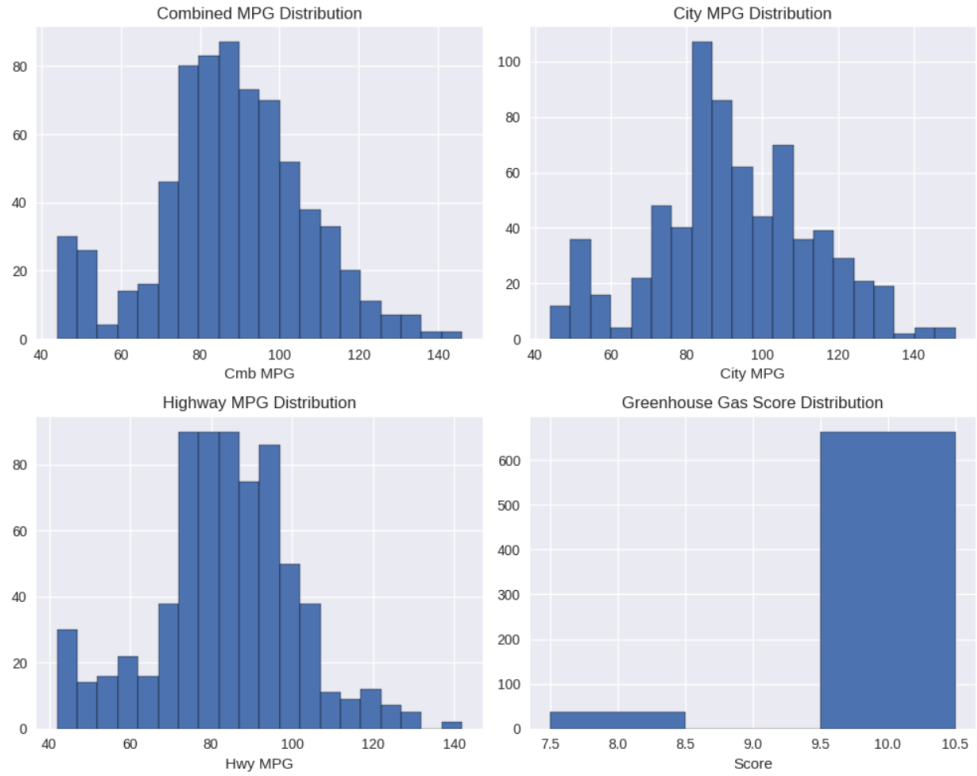

Fig.3: Distributions of Fuel Economy and Emission Scores

Fig.3 examines the univariate distributions of key performance metrics. City, highway, and combined MPG form roughly bell-shaped distributions centered around the mid-80s to low-90s MPG range, with relatively few vehicles achieving extreme low or high efficiency values. The Greenhouse Gas Score distribution is heavily concentrated at the maximum values (mostly scores of 9–10), confirming that SmartWay vehicles consistently achieve strong environmental performance with only a small minority showing lower scores.

4.2. Combined MPG by Fuel Type and Vehicle Class:

Grouped bar charts and boxplots compare performance across categorical factors, highlighting differences in means and variability between groups.

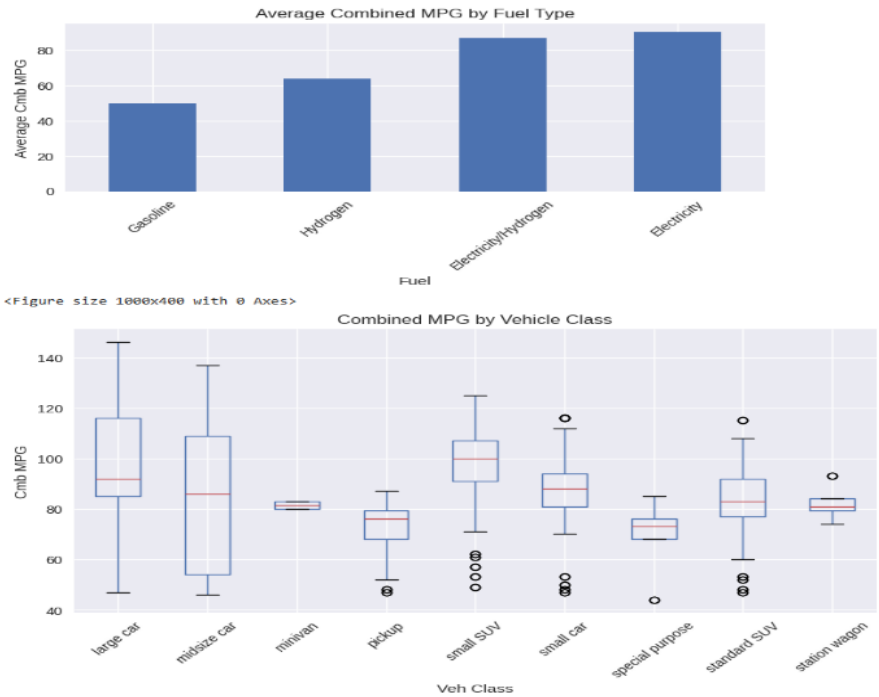

Fig.4: Combined MPG by Fuel Type and Vehicle Class

Fig.4 reveals that electricity vehicles achieve the highest average combined MPG, followed by hydrogen, while conventional gasoline models are noticeably less efficient. The vehicle class boxplots (Fig 4) indicate that small cars and small SUVs tend to have higher combined MPG than pickups and larger SUVs, with large cars and standard SUVs exhibiting wider variability reflecting both efficient and inefficient models within those categories.

4.3. Drivetrain and SmartWay Rating vs Fuel Economy:

Bar charts visualize average performance by certification and configuration factors to assess technology and design impacts.

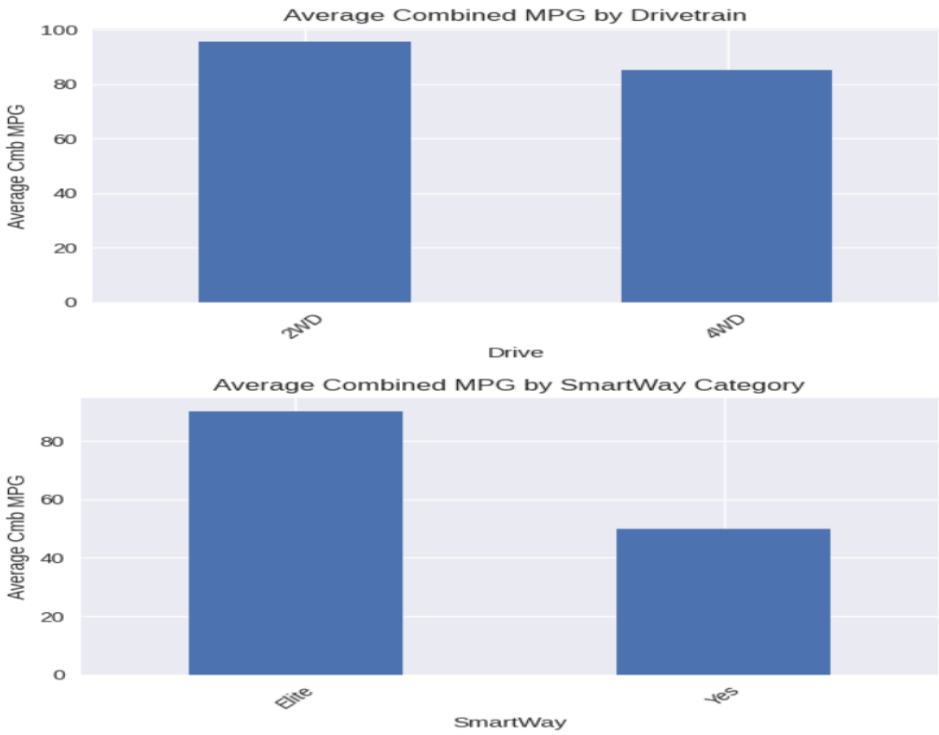

Fig.5: Drivetrain and SmartWay Rating vs Fuel Economy

Fig.5 shows 2WD vehicles achieve higher average combined MPG than 4WD configurations, suggesting four-wheel drive reduces fuel efficiency. Fig 6 confirms that vehicles rated "Elite" by SmartWay substantially outperform those with standard "Yes" ratings, validating the effectiveness of stricter certification criteria.

4.4. Relationship Between Combined MPG and Greenhouse Gas Score:

Scatter plots reveal bivariate relationships between continuous variables to assess correlation strength and direction.

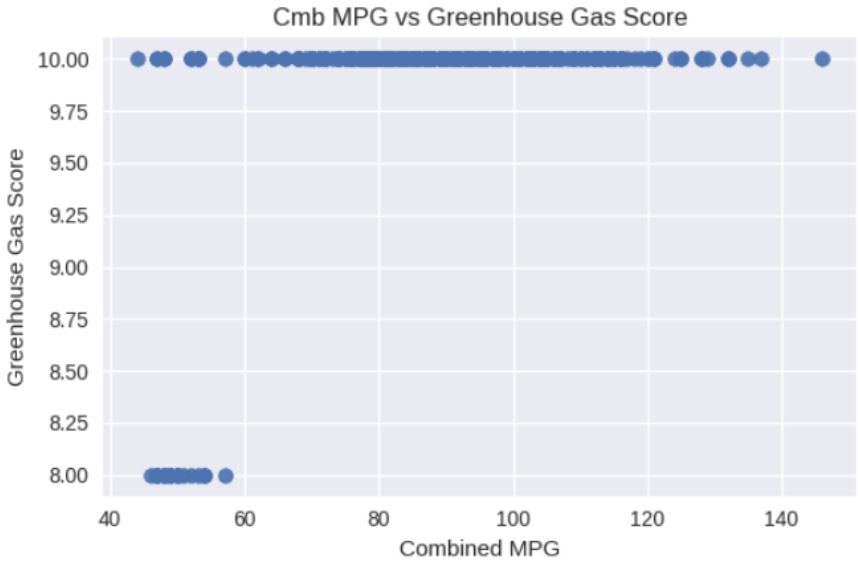

Fig.6: MPG vs Greenhouse Gas Score Relationship

Fig.6 reveals most vehicles cluster at high Greenhouse Gas Scores (~10) across a wide MPG range (40–140), indicating a weak visible correlation between fuel economy and emissions performance among these high-achieving SmartWay models.

5. Statistical Analysis:

5.1. Correlation Between MPG and Emission Scores:

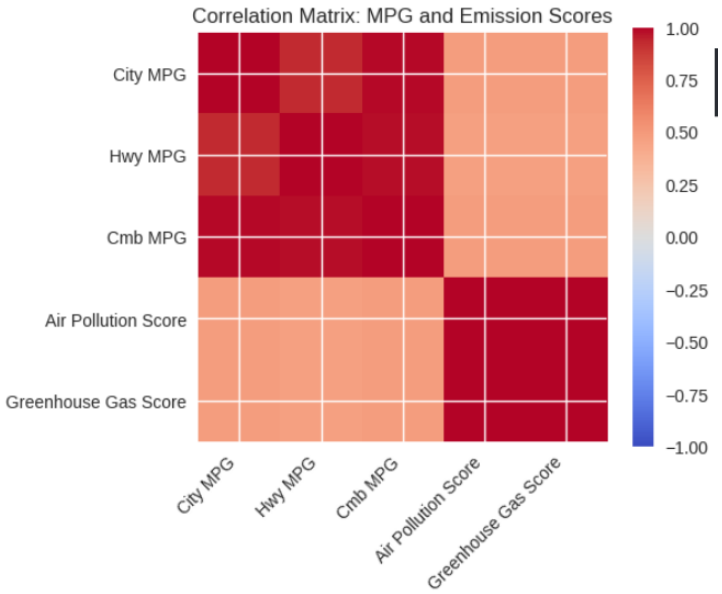

Fig.7: Correlation Between MPG and Emission Scores

Fig.7 presents the correlation matrix for fuel economy (City MPG, Highway MPG, Combined MPG) and emission performance metrics (Air Pollution Score, Greenhouse Gas Score). City, Highway, and Combined MPG exhibit very strong positive correlations (>0.9), indicating vehicles efficient in one driving condition perform consistently across all conditions. Air Pollution Score and Greenhouse Gas Score are strongly correlated with each other but show only weak positive correlations (≈0.2-0.4) with MPG metrics, suggesting fuel economy and emissions represent related but distinct performance dimensions.

5.2. ANOVA: Combined MPG by Fuel Type:

```
from scipy import stats

# Ensure we use only fuels with enough samples
fuel_groups = []
fuel_labels = []

for fuel, group in df.groupby('Fuel'):
    if len(group) > 5:  # simple filter to avoid tiny groups
        fuel_groups.append(group['Cmb MPG'].values)
        fuel_labels.append(fuel)

f_stat, p_val = stats.f_oneway(*fuel_groups)

print("Fuel types included:", fuel_labels)
print("ANOVA - Cmb MPG by Fuel Type")
print("F-statistic:", f_stat)
print("p-value:", p_val)
```

```
Fuel types included: ['Electricity', 'Gasoline', 'Hydrogen']
ANOVA - Cmb MPG by Fuel Type
F-statistic: 114.06011170954122
p-value: 1.42727047164994e-43
```

Fig.8: ANOVA: Combined MPG by Fuel Type

We test the null hypothesis that Electricity, Gasoline, and Hydrogen vehicles have identical mean combined MPG. The one-way ANOVA yields F=114.06 and $p \approx 1.4 \times 10^{-43}$ (Table 1). This extremely low p-value rejects the null hypothesis, confirming at least one fuel type differs significantly in fuel economy. This statistical result validates the visual patterns observed in Fig 4, where electricity vehicles substantially outperform gasoline models.

5.3. ANOVA: Combined MPG by Vehicle Type:

```
veh_groups = []
veh_labels = []

for vcls, group in df.groupby('Veh Class'):
    if len(group) > 5:
        veh_groups.append(group['Cmb MPG'].values)
        veh_labels.append(vcls)

f_stat2, p_val2 = stats.f_oneway(*veh_groups)

print("Vehicle classes included:", veh_labels)
print("ANOVA - Cmb MPG by Vehicle Class")
print("F-statistic:", f_stat2)
print("p-value:", p_val2)
```

```
Vehicle classes included: ['large car', 'midsize car', 'minivan', 'pickup', 'small SUV', 'small car', 'special purpose', 'standard SUV', 'station wagon']
ANOVA - Cmb MPG by Vehicle Class
F-statistic: 20.64133221559221
p-value: 3.6239958607589927e-28
```

Fig.9: ANOVA: Combined MPG by Vehicle Type

In fig.9, the null hypothesis states all vehicle classes have equal mean combined MPG. ANOVA across 9 classes (large car, midsize car, minivan, pickup, small SUV, small car, special purpose, standard SUV, station wagon) produces F=20.64 and $p \approx 3.6 \times 10^{-28}$ (Table 1). We reject the null hypothesis, confirming significant efficiency differences across classes consistent with Fig 4 boxplot patterns showing small vehicles outperforming larger classes.

5.4. T-test: 2WD vs 4WD Drivetrain Efficiency:

```
# If you prefer Drivetrain (2WD vs 4WD), you can instead do a simple t-test:
two_wd = df[df['Drive'].str.contains('2WD', na=False)]['Cmb MPG']
four_wd = df[df['Drive'].str.contains('4WD', na=False)]['Cmb MPG']

t_stat, p_val_drive = stats.ttest_ind(two_wd, four_wd, equal_var=False)
print("T-test - Cmb MPG by Drive (2WD vs 4WD)")
print("t-statistic:", t_stat)
print("p-value:", p_val_drive)
```

```
T-test - Cmb MPG by Drive (2WD vs 4WD)
t-statistic: 5.582984159670581
p-value: 6.21148795493425e-08
```

Fig 10: T-test: 2WD vs 4WD Drivetrain Efficiency

In fig.10, a two-sample t-test compares mean combined MPG between 2WD and 4WD configurations (null: equal means). Results show t=5.58 and $p \approx 6.2 \times 10^{-8}$ (Table 1), rejecting the null hypothesis and confirming 2WD vehicles achieve significantly higher fuel economy than 4WD models, aligning with Fig 5 bar chart observations.

6. Advanced Analysis

Advanced analysis extends beyond descriptive statistics and hypothesis testing to build predictive models that quantify relationships, enable forecasting, and support decision-making under uncertainty. These methods provide interpretable coefficients and model diagnostics for causal inference and scenario planning.

6.1. Linear Regression: Combined MPG vs Greenhouse Gas Score:

Linear regression quantifies the strength and direction of linear relationships between continuous variables, enabling prediction and causal inference while providing interpretable coefficients that reveal effect magnitude

We model Greenhouse Gas Score as a function of Combined MPG using simple linear regression:

$$\text{Greenhouse Gas Score} = \beta_0 + \beta_1(\text{Combined MPG}) + \varepsilon$$

Here, β_0 is the intercept, β_1 is the slope for Combined MPG, and ε is the random error term, with Greenhouse Gas Score as the dependent variable and Combined MPG as the independent variable.

```
x = df['Cmb MPG'].values
y = df['Greenhouse Gas Score'].values

slope, intercept, r_value, p_value_reg, std_err = stats.linregress(x, y)

print("Slope:", slope)
print("Intercept:", intercept)
print("R-squared:", r_value**2)
print("p-value:", p_value_reg)
print("Std err:", std_err)

#Plot
plt.figure(figsize=(6, 4))
plt.scatter(x, y, alpha=0.7, label='Data')
plt.plot(x, intercept + slope * x, color='red', label='Fit line')
plt.xlabel('Combined MPG')
plt.ylabel('Greenhouse Gas Score')
plt.title('Linear Relationship: Cmb MPG vs Greenhouse Gas Score')
plt.legend()
plt.tight_layout()
plt.show()
```

```
Slope: 0.011490633497399335
Intercept: 8.881850180542369
R-squared: 0.23039235539591935
p-value: 1.1153533203155543e-41
Std err: 0.0007943392800730771
```

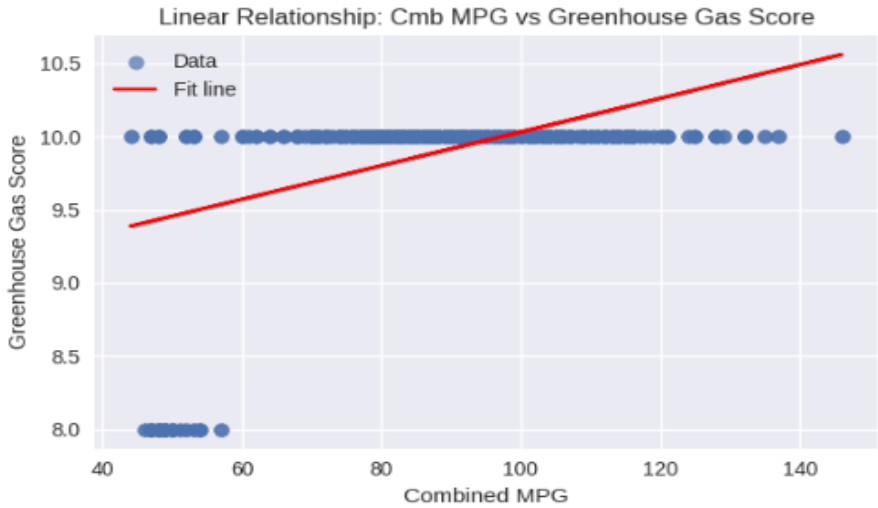

Fig.11: Linear Regression: Combined MPG vs Greenhouse Gas Score

Fig.11 visualizes the regression fit line overlaid on the scatter plot, while Table 1 reports model statistics. The positive slope ($\beta_1 \approx 0.011$) indicates each 1 MPG increase predicts a 0.011-point improvement in Greenhouse Gas Score. The $R^2 = 0.23$ suggests Combined MPG explains 23% of variance in emission scores a moderate but meaningful relationship.

The extremely low p-value ($\approx 1.1\times10^{-41}$) confirms statistical significance; we reject the null hypothesis of zero slope ($\beta_1 = 0$). The standard error (≈ 0.001) indicates precise coefficient estimation

Model Equation:

$$\text{Greenhouse Gas Score} = 8.19 + 0.011(\text{Combined MPG})$$

While higher fuel economy modestly improves greenhouse gas performance, the relatively low R^2 reveals other unmeasured factors (vehicle weight, aerodynamics, manufacturing processes) substantially influence emission scores beyond MPG alone. This finding suggests fuel economy and greenhouse gas ratings capture complementary aspects of environmental performance within the SmartWay program.

Table 1: Statistical Test Summary:

Test	Statistic	p-value	Conclusion
ANOVA: Fuel Type	F=114.06	1.4×10^{-43}	Reject H_0
ANOVA: Vehicle Class	F=20.64	3.6×10^{-28}	Reject H_0
T-test: 2WD vs 4WD	t=5.58	6.2×10^{-8}	Reject H_0
Regression: MPG-GHG	Slope=0.011, R^2=0.23	1.1×10^{-41}	Significant

All tests demonstrate highly significant differences and relationships, providing robust statistical validation of the exploratory visualization patterns**.**

Conclusion: This project selected and documented a unique SmartWay MY 2025 vehicle dataset from data.gov, cleaned and prepared it by converting MPG fields to numeric and handling missing values, and explored it through matplotlib histograms and bar/box plots to reveal efficiency patterns across technologies. We then applied correlation analysis, ANOVA, and t-tests to quantify how fuel type, vehicle class, and drivetrain affect combined MPG, and built a linear regression model to assess how fuel economy predicts greenhouse gas scores. The results show that fuel type is the dominant driver of fuel economy with electricity far outperforming gasoline, while vehicle class and drivetrain play important secondary roles, and that Greenhouse Gas Scores remain consistently high across most vehicles, indicating the SmartWay program's strong environmental effectiveness despite only a modest MPG emission correlation. Overall, the combination of static visualizations and statistical modeling provides clear, actionable insights into vehicle efficiency that can support consumer choice, fleet management, and sustainable transportation policy.